Biology All In One (BAIO) Learning System

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Chapter 9: Biotechnology – Principles and Processes

Worksheet · Selectable Markers & Insertional Inactivation

I. Multiple Choice Questions

1. Genes that help distinguish recombinants from non-recombinants are called:

   (A) Structural genes
   (B) Selectable markers
   (C) Housekeeping genes
   (D) Promoter genes

2. Disruption of a gene's function due to insertion of foreign DNA within it is called:

   (A) Transformation
   (B) Insertional inactivation
   (C) Transduction
   (D) Conjugation

3. Besides antibiotic resistance genes, which gene can serve as an alternate selectable marker?

   (A) β-galactosidase gene
   (B) Haemoglobin gene
   (C) Insulin gene
   (D) Actin gene

4. amp^{R} confers resistance to which antibiotic?

   (A) Tetracycline
   (B) Ampicillin
   (C) Streptomycin
   (D) Penicillin G only

5. tet^{R} confers resistance to which antibiotic?

   (A) Ampicillin
   (B) Tetracycline
   (C) Kanamycin
   (D) Chloramphenicol

II. Fill in the Blanks

1. Disruption of a gene's function due to insertion of foreign DNA within it is called ______.
2. In pBR322, the BamHI recognition site lies within the ______ gene.
3. Besides antibiotic resistance, the ______ gene can serve as an alternate selectable marker.
4. On a chromogenic substrate, non-recombinant colonies (active enzyme) appear ______ in colour.
5. Recombinant colonies with an inactivated marker gene appear ______ on a chromogenic substrate.
6. Recombinants with a disrupted tetR gene grow on ______ medium but not on tetracycline medium.

III. Match the Following

Column A	Column B
1. Blue colony	a. Ampicillin resistance gene
2. Colourless colony	b. Recombinant on chromogenic substrate
3. amp^R	c. Loss of gene function from DNA insertion
4. Insertional inactivation	d. Non-recombinant on chromogenic substrate

IV. True / False

1. Blue colonies on a chromogenic substrate represent successful recombinants. [True / False]

2. Insertional inactivation destroys a gene's original function. [True / False]

3. Non-recombinants lose antibiotic resistance after transformation. [True / False]

4. Recombinants (BamHI insertion in tetR) grow on ampicillin but die on tetracycline. [True / False]

5. Chromogenic substrates help visually distinguish recombinants from non-recombinants. [True / False]

V. 2 Marks Questions

1. Explain how antibiotic resistance genes help distinguish recombinants from non-recombinants.

2. Describe the principle of blue-white screening using the β-galactosidase gene.

VI. 3 Marks Questions

1. Why do non-recombinant colonies grow on both ampicillin and tetracycline media?

2. What is insertional inactivation, and where else besides antibiotic genes can it be applied?

VII. 5 Marks Questions

1. With the help of a diagram, explain how antibiotic resistance genes (ampR and tetR) are used to distinguish recombinant bacteria from non-recombinant bacteria through insertional inactivation.